

Periodic pair inversion and the initial iid endpoint

HOCF–RIESZ, Round 005

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Status at preparation. This memorandum synthesizes the submitted THM–023–025 modules. Independent reconstruction and hostile review are recorded in the canonical theorem ledger. The prescribed-data and homogeneous initial endpoint statements do not establish the singular finite-particle Itô domain, an evolved-law residual estimate, or a fluctuation limit. Complete proofs and immutable audit dossiers are retained in the accompanying Markdown memoranda.

The precise operator and assumptions

The torus has Haar mass one and characters $e^{2\pi i k \cdot x}$. Fix $d \geq 3$, $0 < s \leq d - 2$, $N \geq 2$, $T < \infty$, and $0 \leq \nu \leq \nu_* < \infty$. The frozen periodic Riesz potential has

$$\widehat{g}_s(k) = c_{d,s} |k|^{s-d}, \quad c_{d,s} = \pi^{s-d/2} \frac{\Gamma((d-s)/2)}{\Gamma(s/2)} \quad (k \neq 0), \quad \widehat{g}_s(0) = 0,$$

and $K = -\nabla g_s$. Prescribe a time-continuous C^1 probability density μ_t , a time-continuous uniformly C^1 velocity u_t , and a time-continuous uniformly C^2 test f_t . Write

$$M_0 = \sup_t \|\mu_t\|_\infty, \quad M_1 = \sup_t \|\nabla \mu_t\|_\infty, \quad \operatorname{div} u_t \geq -D_u.$$

All stated norms are uniform on the parameter family in question. Their validity for a general unknown mean-field solution is a separate assertion.

The base pair generator and actual prescribed source are

$$G_{N,t} = \nu(\Delta_x + \Delta_y) + u_t(x) \cdot \nabla_x + u_t(y) \cdot \nabla_y + \frac{1}{N} K(x-y) \cdot (\nabla_x - \nabla_y),$$
$$J_t(x, y) = K(x-y) \cdot (\nabla f_t(x) - \nabla f_t(y)).$$

The full pair operator is $G_{N,t} + \mathcal{R}_{x,t} + \mathcal{R}_{y,t}$. Both responses have coefficient one. Symmetry means commutation with exchanging x, y ; no Haar self-adjointness is asserted.

The base potential and the singular passage

Set $p = s + 2$ and

$$F_{N,\tau}(r) = \frac{N}{4} \left[\left(r^p + \frac{2sp\tau}{N} \right)^{2/p} - r^2 \right].$$

For the local relative generator $2\nu\Delta + (2s/N)r^{-s-2}z \cdot \nabla$, the radial comparison uses

$$\Delta F_{N,\tau} = \frac{N}{2} \left[q^{s/p} (d + s - sq) - d \right] \leq 0, \quad q = \frac{r^p}{r^p + 2sp\tau/N}.$$

The sign holds on precisely the declared range $s \leq d - 2$. In an embedded torus ball, write $K(z) = s|z|^{-s-2}z + k(z)$, where k is smooth and odd. The remaining relative drift has size at most $L|z|$, and $-r\partial_r F \leq sF$. For a fixed radial cutoff χ supported in that ball, the product rule retains the diffusion cross

term $4\nu\nabla\chi \cdot \nabla F$. All cutoff errors are bounded by a constant times τ , uniformly for $N \geq 2$ and $\nu \leq \nu_*$. Consequently constants α, C_B depending only on the prescribed data give the barrier

$$B_N(t, x, y) = e^{\alpha(T-t)} [\|D^2 f\|_\infty \chi(x-y) F_{N,T-t}(|x-y|) + C_B(T-t)], \quad (\partial_t + G_{N,t})B_N \leq -|J_t|.$$

The cutoff is defined in the embedded chart, never by differentiating torus distance across its cut locus.

Local Picard constructions are patched before collisions. The Lyapunov function $1 + \chi(r)r^{-(d-2)}$ has generator bounded above by a constant times itself. Stopped expectation therefore bounds the probability of hitting a tube of radius ε by a constant times ε^{d-2} for each fixed starting point. Compactness excludes other explosion. The global off-diagonal process has deterministic-time Markov conditioning. Stopped Itô applied to B_N , followed by monotone convergence for the nonnegative occupation integral, proves

$$\mathbb{E}_{t,x,y} \int_t^T |J_a(X_a, Y_a)| da \leq B_N(t, x, y).$$

This establishes integrability before defining the signed potential $U_N(t) = \mathbb{E} \int_t^T J_a da$. It is bounded Borel for each fixed N and is the unique terminal-zero bounded Borel martingale potential in the stated class. Radial integration of the barrier, including the finite- N cases where its core exceeds the cutoff radius, yields

$$\sup_t \|U_N(t)\|_{L^2(\mathbb{T}^{2d})}^2 \leq C\rho_N, \quad \rho_N = \begin{cases} 1, & 2s < d, \\ 1 + \log N, & 2s = d, \\ N^{(2s-d)/(s+2)}, & 2s > d. \end{cases}$$

The distribution $D_K = \operatorname{div} K$ is a finite signed measure with $D_K \geq -\kappa$. Choose κ at least as large as the periodic remainder constant C_0 in THM-023. Heat regularization preserves this lower bound. The smooth pair drift has divergence at least $-2D_u - 2\kappa/N$; its random-flow Jacobian, change of variables and Jensen's inequality give

$$\|S_{t,a}^\varepsilon\|_{2 \rightarrow 2} \leq e^{(D_u + \kappa/N)(a-t)}.$$

For each fixed singular path, its minimum separation on a finite horizon is positive. Local C^1 convergence of the regularized force and a stopped Gronwall argument give uniform path convergence. Testing first on continuous terminal functions passes the Haar measure domination to the singular process. Extension to Borel measures and density of continuous functions give the same L^2 bound and strong, fixed-parameter convergence of the semigroups. No operator-norm convergence or convergence of regularized source potentials is inferred from this terminal-data argument.

Restoring both responses

The finite-measure response is

$$\begin{aligned} \mathcal{R}_{x,t}\Phi(x, y) = & - \int \mu_t(x+w)\Phi(x+w, y) D_K(dw) \\ & - \int K(w) \cdot \nabla \mu_t(x+w)\Phi(x+w, y) dw. \end{aligned}$$

The second slot is analogous. Each has Borel supremum and Haar L^2 norm at most $M_0\|D_K\|_{\text{TV}} + M_1\|K\|_1$. At Coulomb,

$$D_K = c_d(\delta_0 - dx), \quad c_d = (d-2)|\mathbb{S}^{d-1}|.$$

The atom contributes $-c_d\mu_t(x)\Phi(x, y)$, not a trace on $x = y$. Off the diagonal, changing diagonal values changes the integral only at $w = y - x \neq 0$, where neither D_K nor the absolutely continuous K term has an atom. Thus the off-diagonal Borel operator is well defined. Haar equivalence-class independence follows separately by translation and Fubini.

Put $C_R = 2(M_0\|D_K\|_{\text{TV}} + M_1\|K\|_1)$ and $c = D_u + \kappa/2$. For

$$(Vh)_t = \int_t^T S_{t,a}(\mathcal{R}_{x,a} + \mathcal{R}_{y,a})h_a da, \quad \Phi = \sum_{j=0}^{\infty} V^j U_N,$$

the ordered time simplex gives a Borel supremum bound $\|U_N\|_\infty (C_R T)^j / j!$. The intervening semigroup exponents add over adjacent time intervals, so the L^2 bound is $e^{cT} \sup_a \|U_N(a)\|_2 (C_R T)^j / j!$. Both series converge consistently; the same factorial estimate proves uniqueness in the two separate classes. In particular

$$\sup_t \|\Phi_t\|_2 \leq e^{(c+C_R)T} \sup_t \|U_N(t)\|_2.$$

Conditional Markov time splitting identifies the solution with the true uniformly integrable martingale whose integrated source is $J + (\mathcal{R}_x + \mathcal{R}_y)\Phi$. Absolute occupation integrability justifies the time interchanges. This is a probabilistic inverse; it is not a singular Itô formula for the N -particle corrector.

The initial endpoint and an actual homogeneous subcase

For iid initial law μ_0 , retain the exact mean-field-centered statistic $P_N = U_2/2$ with denominator N^2 . THM–015 gives, with $b_N = \min(\beta_N, 1)$,

$$\begin{aligned} \mathbb{E} \left| \sqrt{N b_N} P_N[\Phi_0] \right|^2 &\leq \frac{b_N}{2N} \|\Phi_0\|_{L^2(\mu_0^2)}^2 \\ &\leq \frac{b_N M_0^2}{2N} e^{2(c+C_R)T} C \rho_N. \end{aligned}$$

The factor is M_0^2 : the full kernel need not depend only on $x - y$. The deterministic mean and first projection have not been omitted. The rates are b_N/N , $b_N(1 + \log N)/N$, and $b_N N^{-(d+2-s)/(s+2)}$, respectively. These all vanish in the declared range.

For actual homogeneous data, take $b = 0$, $\mu_0 = 1$. Then $\mu_t = 1$, $u_t = 0$ solve the reference equation. For a real smooth terminal test h , set

$$f_t(x) = \widehat{h}(0) + \sum_{k \neq 0} \widehat{h}(k) e^{-(T-t)a_k(\nu)} e^{2\pi i k \cdot x}, \quad a_k(\nu) = 4\pi^2 \nu |k|^2 + 4\pi^2 c_{d,s} |k|^{s+2-d}.$$

All a_k are positive. Rapid Fourier decay justifies time and spatial differentiation and bounds every fixed spatial derivative uniformly for $t \in [0, T]$ and $\nu \geq 0$. The one-body response multiplier is $-4\pi^2 |k|^2 \widehat{g}_s(k)$; hence $\partial_t f + \nu \Delta f + \mathcal{R}f = 0$ with $f_T = h$. This verifies the required background and test norms for this explicit actual-data subcase. The pair estimate still requires bounded ν .

With $\nu = 1/\beta_N$, the conclusion therefore covers $\beta_N \geq \beta_* > 0$. It includes every microscopically critical sequence $\lambda_N = \beta_N N^{s/d-1} \rightarrow \lambda \in (0, \infty)$, which eventually has $\beta_N \rightarrow \infty$, and subcritical sequences satisfying the lower bound. Subcritical sequences with $\beta_N \rightarrow 0$ are not covered. The old condition $\beta_N N^{2s/d-1} \rightarrow 0$ remains distinct from microscopic subcriticality.

First remaining line. The constructed pair kernel is not yet shown to have the gradients and approximation properties needed in the interacting finite-particle identity. Its cubic residual, lower contractions and self/cross brackets must then be controlled under the evolved law. Neither the initial iid estimate nor the auxiliary martingale establishes those statements; critical hierarchy tails also remain open.